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SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Failure Mechanisms and Detection Parameters (1 of 4)

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Failure Mechanisms and Detection Parameters

Learning Objectives

By the end of this session, you should be able to:

- Understand the various degradation mechanisms affecting equipment components i.e. RCFA approach
- Understand the failure physics relating the degradation parameters for specific failure mechanism i.e. the deterministic approach, to be used in RCFA and detecting incipient failures through Predictive Maintenance Techniques.
- Aware of the re-engineering and RCFA approach that can be implemented to design out or avoid recurrence of failure.
- Aware to the various technology for detection of degradation parameters, and interpretation of results.

Some Definitions

Term	Definition
Criticality	The <u>relative severity</u> of a piece of equipment in the operating conditions.
Failure	Termination of the ability of an item to perform a required function.
Failure Mode	Failure Mode is the effect by which a <u>failure is observed</u> on the failed item^{1,2}. ¹ISO 14224, second edition, 2006, page 4 and 120 ²Hidden (observed during demand/start-up), or evident (observed during <i>operation</i>) failures.
Failure Mechanism	Failure Mechanism is the <u>physical, chemical or other process or combination of processes that leads to the failure</u>. It is an <u>attribute of the failure event that can be deduced (and validated²) technically</u>, e.g. the apparent, observed cause of the failure³. ²Added by RIMS. ³ISO 14224, second edition, 2006, page 114.

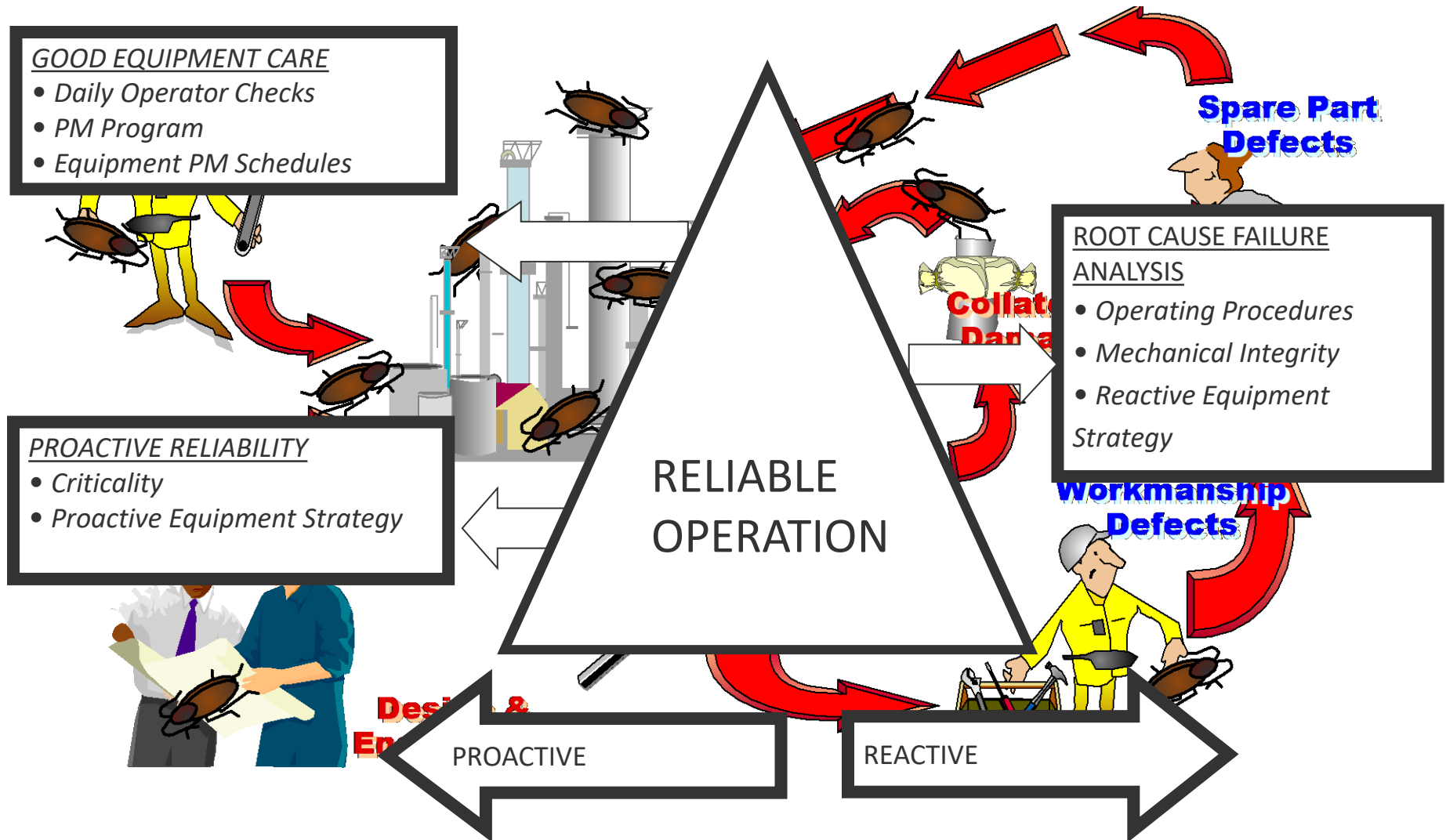
Some Definitions

Term	Definition
Consequence of Failure	The consequence of a failure measured in terms of Health & Safety, Production Loss (economics) and Environment.
Critical Detection or Operating Parameter (COD/COP)	is defined as process and equipment parameters that at some measurable level indicates an approach to undesirable event or MAH. COP / CODs are used as the basis for the controls that need to be in place for reducing consequence or Failure or risk of MAH. The COPs should be identified based on:
Parameter should be M onitorable, C ontrollable and A uditable	• Operating / Maintenance Procedures, FMEAs, Recommendations by Process Licensor or OEMs, or • Major risks as identified in Hazard Effect Register (HER), • Parameters that lead to MAH which solely depends on human intervention to bring it back to safe operation and with or without any automatic protective system (IPF, Vibration, Full Load Current, Voltage, Temperature, CBM and other Process Parameters)

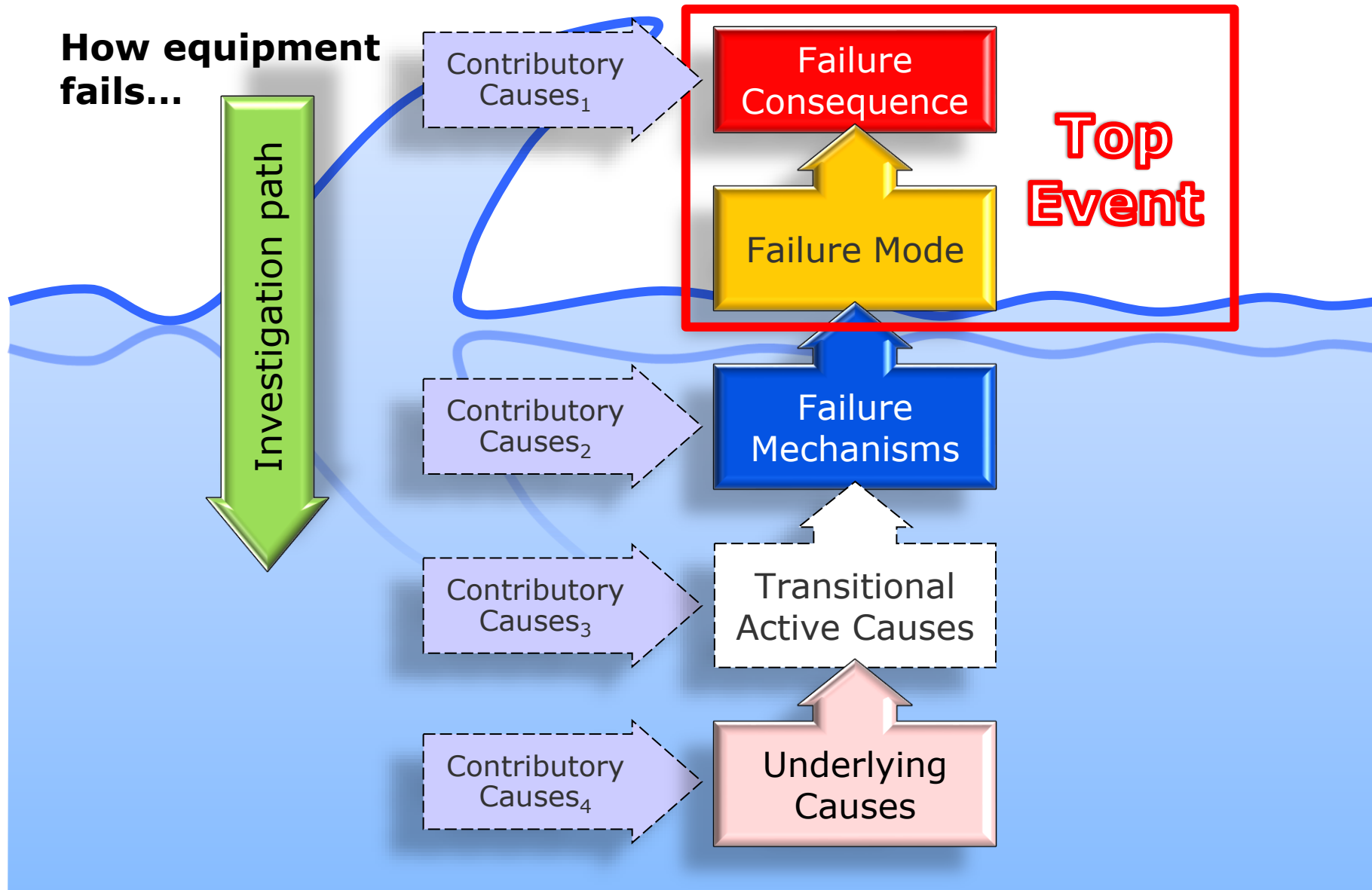
Some Definitions

Term	Definition
Failure Mechanism Reference and Application	Reference is made to: • The information related to equipment degradation mechanisms (EDM's) • Their probability of Failure • The tasks to be applied for mitigation including Equipment EDMs and Process EDM's(viz fouling, catalyst deactivation). Application: • list of possible threats# (EDM's) that equipment can suffer and use past history, symptoms, operating conditions to diagnose possible cause and develop remedial / improvement actions. • It's easier, less expensive, and healthier (safer) to prevent threats (EDM's) than it is to cure them. # Preventing vibration, fatigue, cracks, leaks, ruptures, high corrosion rates, and metallurgical degradation is usually easier, much less expensive, and safer than coping with the aftermath of equipment failures

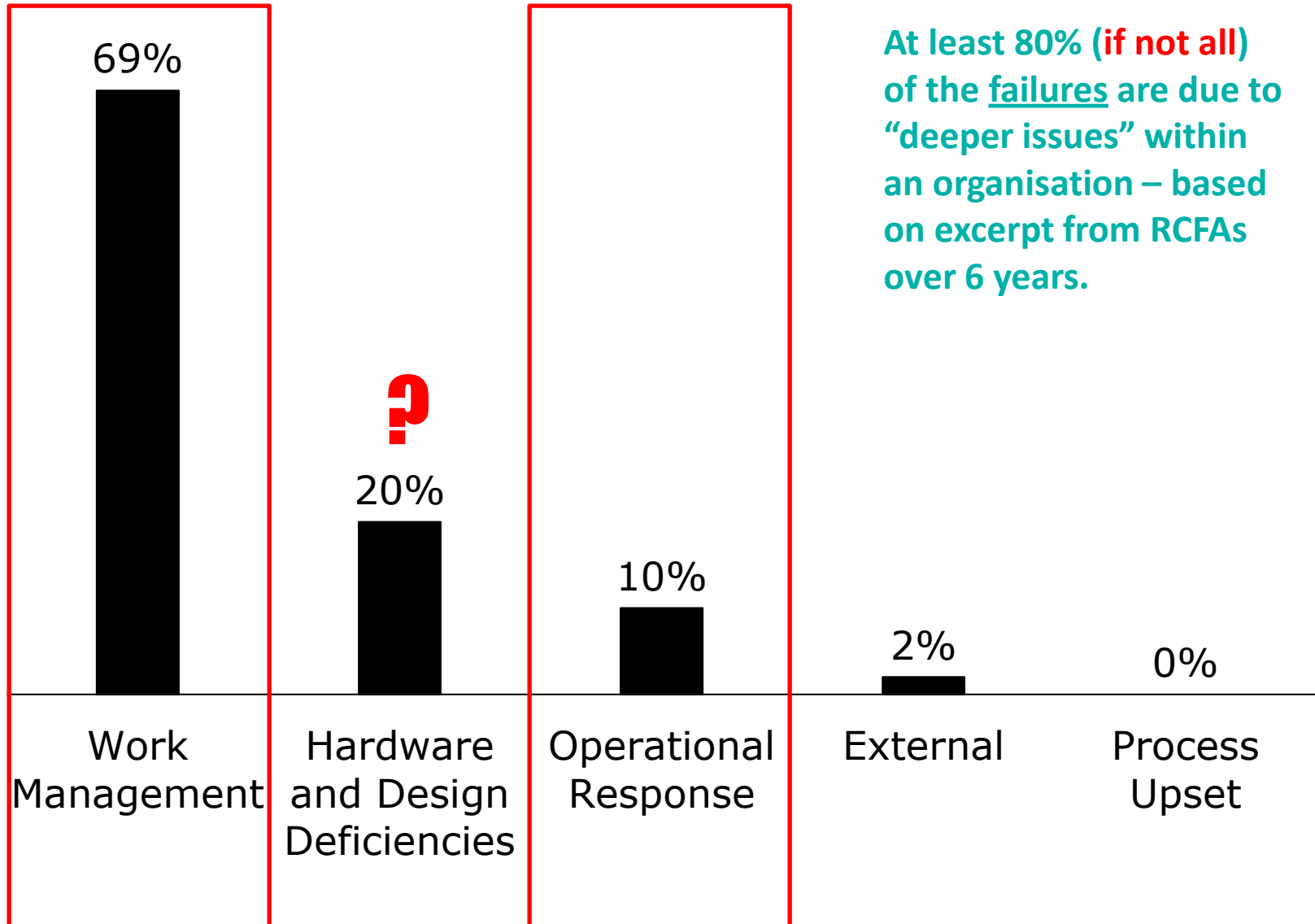
The Way We Look At Vicious Cycle of Typical Operating System with Defects / Bugs



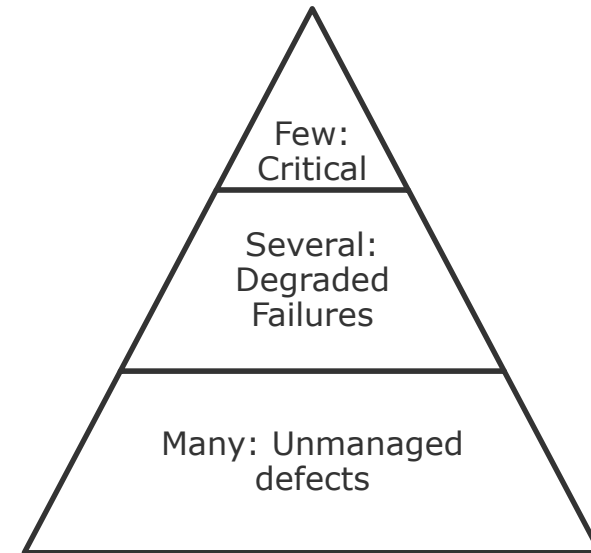
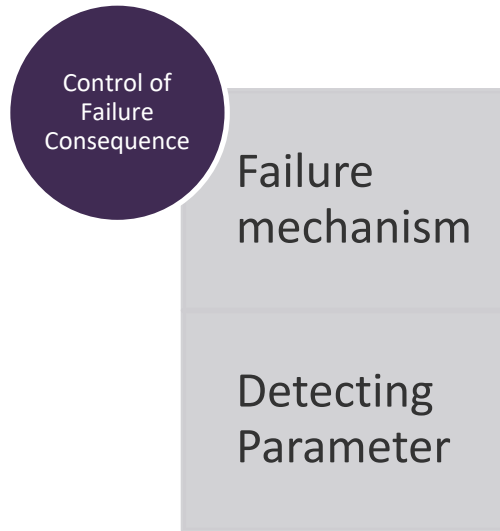
Background on Failure and its Analysis



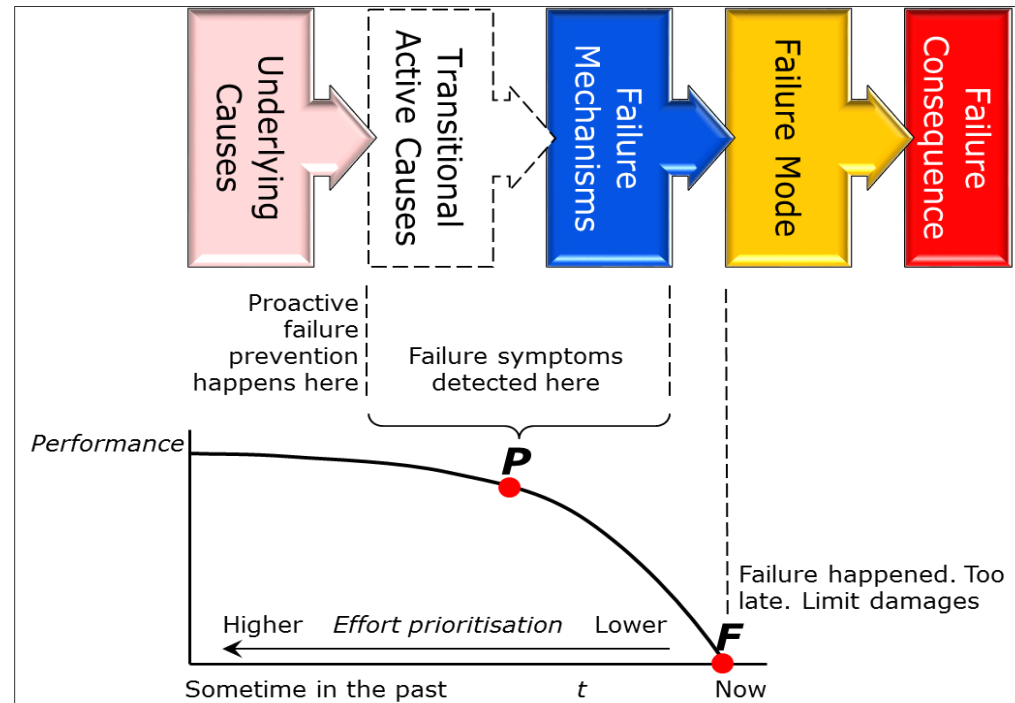
Background on Failure and its Analysis



Background on Failure and its Analysis



- Failures (and hence the consequences) happen due to the unmanaged defects – very rarely by chance.
- Not ALL Defects are Failures; but they can lead to failures
- Addressing those early, will be much easier.
- These defects or symptoms could be identified as part of TAC or FM.
- Hence, understanding of FM, its detecting Parameter and continuous pursuit of sources of defects is vital.



Typical Monitoring or Detection Parameters

INDICATIVE LIST OF PLANT EQUIPMENT FOR DETECTION PURPOSES

MECHANICAL

Cranes
Heat exchangers
/Reboiler
Feed Heaters

Vessels

Piping
FURNACE
COLUMNS
COALESCER
Storage Tanks
Filters and strainers
Ejectors / Drier
INCINERATOR
Surge Drum
Reactors
Product / Overhead
Cooler
Regenerators
BOILER (Fired / WH)

ELECTRICAL

UPS

Power transformers

Switchgears

Electric motors

Earthing System LPS
HVAC
Emergency Generator
Alternator
Variable Speed
Cathodic Protection
Motor Operated Valve

INST&CONT

Initiators

Controller / Sensor

Control System

Shutdown System

Metering
Analyzer
Speed Sensor
Vibration Sensor
Interlock system
Shutdown Valve
Blowdown Valve
HIIPS
Fire & Gas Detectors

ROTARY

Product Pump

Circulation Pump

Feed Pump

Recycle Gas Compressor

FG Compressor
MUG Compressor
Steam Turbine
Gas Turbine
Expander
Blowers and fans
Combustion engines
Mixers and Agitators

SAFETY

Deluge System

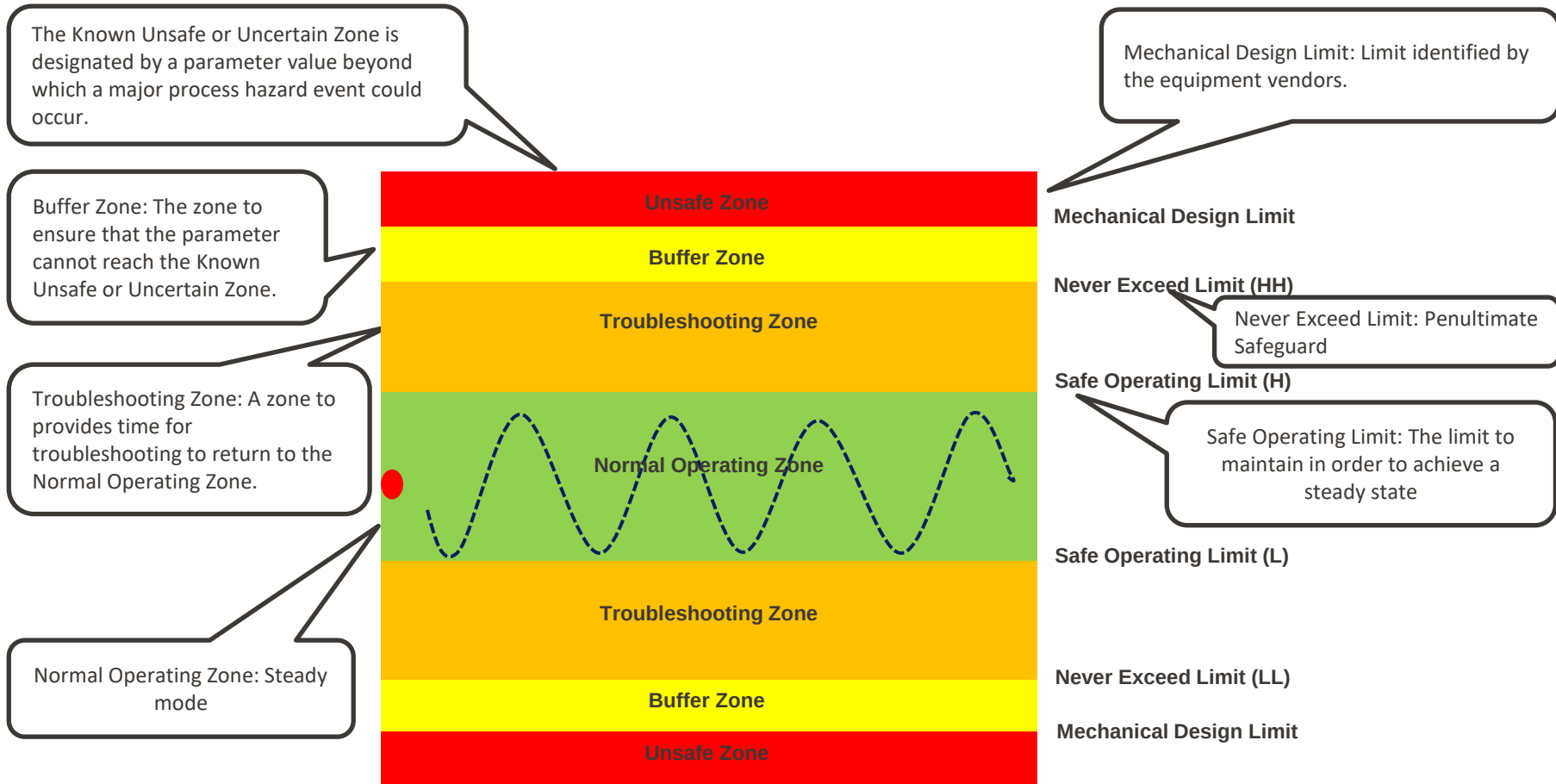
Gas Suppression System

Inert gas System
Personal Survival
Equipment
Fire Water Pump
Fire Call Points

TYPICAL PARAMETERS

PARAMETER	UNITS
Pressure	kPag
Amperage	amp
Quality	ppm Wt
Flow	m3/hr
Temp	C
Level	%
Vibration	µm
Speed	RPM
Efficiency	%
Thickness	mm

Operating or Detection Parameter Target Zone



In this module topics covered

Failure Mechanisms and detection parameters for:

- Static equipment (Part 2 of 4)
- Rotary equipment (Part 3 of 4)
- Electrical and Instrument Control equipment (Part 4 of 4)



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Thank you